VANGUARD

One picture. Every source. Zero delay.

Pitch Deck & Presentation

Multi-Source Defence Situational Awareness System
Problem ID D-05 · HackHertz 2026 Defense Track
Team Destroyer of Worlds · Document version 1.1

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14 slides · 6 minutes · one live demo

Slide 1 — Title

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Multi-Source Defence Situational Awareness System

Problem D-05 · Defense Track Team Destroyer of Worlds

> *Speaker note (10s):* Open on the live command center already running. Let the map move behind you while you say the tagline. Do not read the slide.

Slide 2 — The problem

A watchstander has five screens and ten seconds

Feed	Sits in	Says
Radar	Its own console	"Contact, 23.02N 72.57E, 260 knots, no transponder"
Perimeter sensors	A different console	"IR trip, sector 4, amplitude 0.87"
Patrol telemetry	A tablet	"GRIZZLY-1 reports visual contact"
Weather	A web terminal	"Visibility 1.8 km"
Dispatch	A radio net	"Unauthorized entry, sector 4"

Five systems. One event. Nobody sees it.

> *Speaker note (35s):* This is the whole pitch. Five different systems just described the **same intrusion** and not one of them knows the others exist. The officer has to assemble that in their head, under time pressure, while the situation develops.

Slide 3 — Why "put it on one screen" is not the answer

Aggregation makes it worse

One screen with five times the noise is not situational awareness.

Fusion asks a harder question:

> *Which of these observations are about the same thing?*

Four hard problems underneath:

1. **Correspondence** — no shared ID links a radar return to a sensor trip
2. **Conflict** — radar holds a contact; the patrol there sees nothing
3. **Variable reliability** — a calibrated radar ≠ an anonymous tip
4. **False corroboration** — one radar reporting six times is **not** six confirmations

> *Speaker note (30s):* Point 4 is the one that kills naive systems. Redundancy dressed as independence produces confident wrong answers — worse than no fusion at all.

Slide 4 — Why generic AI fails here

Failure	Looks like	Why it disqualifies the tool
Fabricated specifics	*"Three contacts at 240 knots"* — there were two at 180	An officer may act on it
Unfalsifiable confidence	*"High confidence of a breach"*	Where did that come from?
Invented citations	Cites event IDs that don't exist	Looks **more** rigorous while being less

An unverifiable summary is worse than no summary

It carries the authority of a system without the accountability of one.

> *Speaker note (25s):* Defense judges have seen a dozen ChatGPT wrappers today. Name the failure mode before they do.

Slide 5 — VANGUARD

The fusion engine is the product.

The language model is a presentation layer.

```
5 feeds → normalize → DEDUPE → CORRELATE → CORROBORATE
                    → SCORE → ANOMALY → ESCALATE → posture
                                   ↓
                        AI phrases the finished analysis
                                   ↓
                          GROUNDING GATE (code, not prompt)
```

Every number is deterministic arithmetic. The model never produces one.

> *Speaker note (25s):* This inverts the usual hackathon architecture and it is the single most important slide. Everything else follows from it.

Slide 6 — Live: five feeds, one contact

(Demo — click a corroborated contact on the map)

```
EV-RAD-T-R-1012   Uncorrelated uav contact          CRITICAL   100%
  corroborated by:
    EV-LOG-000031  Perimeter trip P-302      430m away, 12s apart
    EV-PER-000037  Visual — GRIZZLY-1        890m away, 31s apart
    EV-INC-000024  Unauthorized entry       1.2km away, 48s apart
```

Three independent instruments. One physical intrusion. Found automatically.

> *Speaker note (30s):* This is the money shot. Say the sentence: *"Nobody told the system these were related. It worked it out from space, time, and source physics."*

Slide 7 — Explainable confidence

$\text{Confidence} = \min(100,\ R_s \times D_t \times B_c \times 100)$

Factor	Value	Meaning
Source reliability	0.92	Radar, healthy
Recency decay	0.999	1.2 seconds old
Corroboration boost	1.60	4 distinct sources (capped)

The number that matters is a subtraction

```
with corroboration:     100%
without corroboration:   92%
              fusion:    +8
```

> *Speaker note (35s):* Open the Explainability Drawer live. The counterfactual converts an abstract claim into a measured quantity — *this is what fusion bought you.*
> **The cap on the boost is deliberate:** it is what stops correlated noise manufacturing certainty.

Slide 8 — The anti-hallucination guarantee

We assume the model is lying, and check

```
1. Resolve every supportingEventId against the store
2. STRIP  every ID that does not resolve
3. DISCARD every claim left with zero citations
4. RECOMPUTE overall confidence from survivors
```

Enforced in code (`ai/grounding.ts`), not requested in a prompt.

```
POST /api/v1/ai/verify
→ { "verified": true, "totalCitations": 42, "ungroundedCitations": [] }
```

> *Speaker note (30s):* Prompting is a request, not a constraint. Run the verify endpoint live. Invite a judge to name any citation and open it.

Slide 9 — Live: escalation

(Demo — trigger `border_spike`)

```
7 correlated entry reports over 3.5km  +  2 non-squawking fast contacts

  rate anomaly       → incident feed 3.2σ above its own baseline
  kinematic anomaly  → 312kt against a 92kt mean (3.0σ)
  correlation        → multi-source cluster forms
  escalation         → 3 distinct sources → severity +1 tier

  THREAT   ORANGE (70.9)  →  RED (147.0 → 221.9)
  CRITICAL      2         →       10
```

> *Speaker note (30s):* One operator action drives the entire pipeline end to end. Say the anomaly numbers out loud — they are real, from the running system.

Slide 10 — Live: resilience

(Demo — engage degraded comms)

```
                    BEFORE      DEGRADED      RESTORED
mean confidence      93%          43%           92%
feed reliability     0.92         0.37          0.92
```

Degradation propagates into the math, not into a banner

Feed health multiplies directly into the confidence formula. When a feed degrades, **every score across the picture falls automatically.**

> *Speaker note (25s):* Most systems paint an amber dot. Ours lowers the trust weight of everything that feed reports. That is the difference between displaying a problem and modelling one.

Slide 11 — The proof: delete the API key

(Demo — regenerate the briefing with no key)

```
{
  "headline": "10 critical events active — Uncorrelated uav contact R-1012",
  "keyDevelopments": [ /* 4, all cited */ ],
  "coursesOfAction": [ /* 3, with honest tradeoffs */ ],
  "provenance": { "engine": "deterministic", "citationsStripped": 0 }
}
```

Full briefing. Real citations. No model call.

This is why it is not a wrapper.

> *Speaker note (30s):* The strongest 30 seconds available. Do not skip it. Then: *"The LLM improves the prose. The fusion engine supplies the substance."*

Slide 12 — Engineering

Tests	114 passing · strict TypeScript · zero `any`
Fusion pass	2–37 ms against a 3000 ms tick budget (~50× headroom)
Correlation	Grid-indexed union-find — 803 comparisons for 118 events vs ~7,000 naive
NL omnibar	0–2 ms, no API call, works offline
Dependencies	4 runtime: `express`, `ws`, `cors`, `dotenv`
Setup	`npm install && npm run dev` — no key, no DB, no Docker

Seven real bugs found by running it, each with a regression test

Compounding escalation · entity duplication · cluster over-escalation · uncorroborated CRITICAL promotion · rule chaining · threat miscalibration · empty feed at boot

> *Speaker note (25s):* The bug list is a credibility asset, not an admission. It says we ran this, measured it, and fixed what we found.

Slide 13 — Evaluation alignment

Criterion	Weight	What we built
Data fusion & multi-source integration	30%	5 feeds incl. **live** Open-Meteo · `UnifiedEvent` normalization · haversine+temporal correlation · explainable confidence · 3 anomaly detectors · health→reliability coupling
Command map & geospatial UX	25%	MapLibre GL · 4 GeoJSON layers · clustering · severity-weighted heatmap · 4D time-scrubber via `snapshotAt`
AI summarization & prioritization	25%	Grounded briefings · **enforced** citation checking · ranked COAs with tradeoffs · NL omnibar · explainability drawer
Scalability & craftsmanship	20%	114 tests · 50× perf headroom · dual-mode fallback · degraded-comms simulation · zero-friction setup

Slide 14 — Close

VANGUARD

One picture. Every source. Zero delay.

What we actually built: A deterministic multi-source fusion engine whose every number can be checked by hand, with an AI layer that cannot lie to you because the code will not let it.

Purely defensive. No kinetic targeting. No weapons release. No lethal autonomy.

```
github.com/Destroyerved/Vanguard
```

> *Speaker note (20s):* End on the running system, not the slide. Offer: *"Name any number on that screen and I will show you where it came from."*

Timing

Slides	Content	Time
1–4	Problem framing	1:40
5	The thesis	0:25
6–7	**Live: fusion + explainability**	1:05
8	Grounding guarantee	0:30
9–11	**Live: escalation, resilience, no-key proof**	1:25
12–14	Engineering, alignment, close	1:10
	Total	**~6:15**

If cut to 4 minutes: keep 1, 2, 5, 6, 7, 11, 14. The demo beats the slides every time.